BODY POSE CONDITIONING IMPROVES PEDESTRIAN TRAJECTORY FORECASTING: GEODESIC ROTATION SUPERVISION AND TEMPORAL ENCODING ARE KEY

Anonymous authors
Paper under double-blind review

ABSTRACT

Pedestrian trajectory forecasting in autonomous driving relies almost exclusively on observed positions, ignoring the rich body pose information visible in modern perception stacks. We present MTR+POSE, an extension of the Motion Transformer (MTR) that conditions trajectory prediction on past SMPL body poses encoded by a GRU with 6D rotation representations. Across a controlled ablation study on the Waymo Open Motion Dataset, we find that pose conditioning reduces minADE by **7.6%** (0.6745 $\rightarrow$ 0.6231) over a trajectory-only baseline—but only when the pose encoder is supervised with the *geodesic distance* in SO(3), not with the commonly used joint-position MPJPE. Combining MPJPE with geodesic loss *hurts* performance, while geodesic loss alone outperforms all other configurations. We further ablate the temporal encoding architecture: cross-attention *without* positional encoding provides essentially no improvement (0.6765 vs. baseline 0.6745), while cross-attention *with* sinusoidal positional encoding recovers 83% of the GRU's advantage (0.6337, -6.1%). A GRU encoder, which accumulates a running hidden state through sequential frames, achieves the full 7.6% gain (0.6231). These results reveal two design principles: (1) rotation-space supervision is more informative than joint-space supervision for encoding gait; and (2) temporal ordering of pose frames is the dominant requirement—sinusoidal PE recovers most of the benefit—while the GRU's causal sequential integration provides a further incremental gain.

1 INTRODUCTION

Predicting where pedestrians will walk is a safety-critical problem in autonomous driving. State-of-the-art trajectory forecasting models such as MTR (Shi et al., 2022; 2024) achieve impressive accuracy by operating on observed agent positions, velocities, and HD map geometry—but they discard a potentially rich source of information: the *body pose* of the pedestrian.

Human gait encodes intent. A pedestrian who is turning to face a crosswalk will likely cross it; one whose torso is oriented away from traffic is unlikely to step into the road. A runner's body orientation predicts turns before the feet complete them. Modern perception systems increasingly provide body pose estimates (Zhu et al., 2023), yet trajectory forecasting pipelines ignore these estimates entirely. This paper asks: *does conditioning on past SMPL body poses improve pedestrian trajectory prediction, and if so, how?*

We extend MTR with a *pose conditioning branch* that encodes the past 11 timesteps of SMPL body pose (Loper et al., 2015) using 6D rotation representations (Zhou et al., 2019), a GRU encoder, and a multi-layer pose prediction head. We train and evaluate on a pedestrian subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with AMASS-derived body poses (Mahmood et al., 2019), using the standard minADE and minFDE metrics over 6 predicted modes.

Our experiments yield three principal findings:

- **Pose conditioning improves trajectory prediction by 7.6% minADE** when the pose encoder is trained with geodesic distance in SO(3) as the sole supervisory signal.

- **Geodesic supervision is the critical ingredient**. Using MPJPE (joint-position L1) instead of geodesic loss degrades accuracy relative to geodesic-only; combining both losses *hurts* compared to geodesic alone. Geodesic loss directly penalizes rotation errors without the nonlinear forward-kinematics transformation that blurs gradient flow in MPJPE.

- **Temporal ordering dominates; sequential integration adds incrementally**. Cross-attention *without* positional encoding yields no improvement (0.6765 $\approx$ baseline 0.6745). Adding sinusoidal positional encoding recovers 83% of the GRU's advantage (0.6337, -6.1%). The GRU's causal sequential integration achieves the full -7.6% gain (0.6231), providing a further 1.7% over cross-attention with PE. Temporal ordering is the dominant requirement; the GRU's running hidden state adds an incremental benefit from causal accumulation.

A fourth finding clarifies the mechanism: using 30fps AMASS pseudo-ground-truth future poses (99% step coverage vs. zero future supervision) gives essentially identical trajectory accuracy (0.6567 vs. 0.6532), confirming that the benefit arises entirely from the *past* pose encoder, not from future pose prediction quality.

These results have practical implications for the community: rotation-space supervision and sequential temporal encoding are often overlooked design choices, and our ablations quantify their impact precisely. Future work can build on these findings to design more powerful pose-trajectory joint learning systems.

2 Related Work

2.1 Trajectory Forecasting for Autonomous Driving

Modern trajectory prediction methods model multi-agent interactions over HD map features using transformer architectures (Vaswani et al., 2017; Shi et al., 2022; 2024; Yuan et al., 2021; Salzmann et al., 2020). Social GAN (Gupta et al., 2018) introduced generative multimodal prediction for pedestrians; MTR (Shi et al., 2022) advances this with global intention localization and local refinement via cross-attention over map and agent context. None of these methods incorporate body pose information.

2.2 Human Pose Estimation and Pose Representations

SMPL (Loper et al., 2015) provides a parametric body model where each joint rotation is represented as a 3D rotation matrix. For neural network learning, Zhou et al. (2019) demonstrate that 6D rotation representations (the first two columns of $SO(3)$) are continuous and better conditioned than quaternions or axis-angle, particularly under gradient-based optimization. The AMASS dataset (Mahmood et al., 2019) provides a large-scale archive of motion-captured body animations in SMPL parameterization, enabling pseudo-ground-truth future poses for pedestrian sequences. MotionBERT (Zhu et al., 2023) learns unified human motion representations from video, while HumanMAC (Chen et al., 2023) addresses probabilistic human motion prediction. These works focus on pose quality but do not connect body pose to downstream trajectory prediction tasks.

2.3 Joint Pose and Trajectory Models

Prior work has linked body pose to pedestrian intent in controlled settings (Gupta et al., 2018), but to our knowledge no existing work jointly models SMPL full-body pose and trajectory prediction end-to-end within a state-of-the-art transformer forecasting framework. Our work fills this gap and contributes a systematic ablation that isolates the contributions of pose supervision type, pose weight, temporal encoding, and future pose quality.

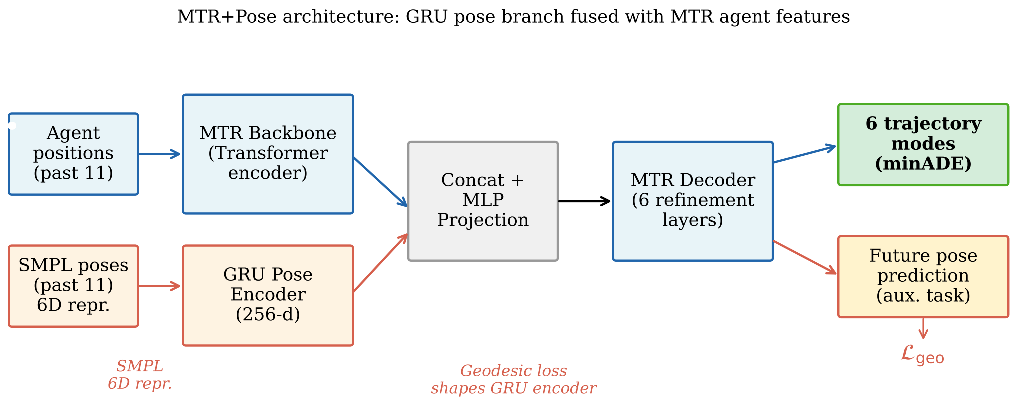

Figure 1: **MTR+Pose architecture.** A GRU pose encoder processes the past 11 SMPL body poses (each represented as a 144D 6D-rotation vector) and produces a 256D hidden state that is concatenated with the MTR agent feature and projected before the decoder. Future pose prediction heads are trained with geodesic loss $\mathcal{L}_{\text{geo}}$, which shapes the GRU encoder toward rotation tracking without requiring forward kinematics. The trajectory decoder and map attention modules of MTR are unchanged.

3 METHOD

3.1 BACKGROUND: MOTION TRANSFORMER (MTR)

MTR (Shi et al., 2022) is a transformer-based trajectory predictor for the Waymo Open Motion Dataset. For each *center object* (the agent being predicted), it encodes a scene context tensor from agent histories and HD map polylines via a PointNet-like encoder, then decodes $K = 6$ trajectory modes through iterative cross-attention over the scene context, guided by K learned intention queries that are globally localized to trajectory clusters.

The loss has three components: a mode classification loss $\mathcal{L}_{\text{cls}}$, a winner-takes-all regression loss $\mathcal{L}_{\text{reg}}$, and a velocity consistency loss $\mathcal{L}_{\text{vel}}$.

3.2 POSE CONDITIONING BRANCH

We add a pose conditioning branch alongside the trajectory decoder, illustrated in Figure 1. The branch has four components:

6D Rotation Encoding. Each SMPL body pose at timestep t consists of 24 joint rotations. We represent each rotation as a 6D vector (Zhou et al., 2019) (the first two columns of the 3×3 rotation matrix), giving a 144D pose vector per timestep.

GRU Pose Encoder. Given the past $T = 11$ pose vectors $\mathbf{p}_1, \ldots, \mathbf{p}_T \in \mathbb{R}^{144}$, a single-layer GRU with hidden size 256 processes the sequence:

$$\mathbf{h}_t = \mathrm{GRU}(\mathbf{p}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_T \in \mathbb{R}^{256}. \tag{1}$$

The final hidden state $\mathbf{h}_T$ is concatenated with the MTR agent feature and projected to 256D via an MLP, forming the fused agent representation.

Pose Prediction Heads. Each of the K decoder layers predicts future poses. At layer ℓ and mode k, a 144D linear head predicts 6D rotations for future timesteps. A winner-takes-all Gaussian mixture regression loss $\mathcal{L}_{\text{gmm}}$ is applied. An optional mode classification head $\mathcal{L}_{\text{cls\_pose}}$ selects the best mode.

Pose Supervision Loss. Let $\hat{\mathbf{r}}^{ij} \in \mathbb{R}^6$ be the predicted 6D rotation and $\mathbf{r}^{ij} \in \mathbb{R}^6$ be the ground-truth 6D rotation for joint j at future step i. We consider three supervision signals:

- **Geodesic loss**: $\mathcal{L}_{\text{geo}} = \frac{1}{IJ}\sum_{i,j}\arccos\Big(\text{clamp}\Big(\frac{\text{tr}(\mathbf{R}_i^{j\top}\hat{\mathbf{R}}_i^j)-1}{2},\ -1+\epsilon,\ 1-\epsilon\Big)\Big)$, where $\mathbf{R}, \hat{\mathbf{R}} \in SO(3)$ are the full rotation matrices recovered from 6D representations. Gradients vanish at neither 0 nor π.
- **MPJPE**: $\mathcal{L}_{\text{mpjpe}} = \frac{1}{IJ}\sum_{i,j}\|\mathbf{x}_i^j - \hat{\mathbf{x}}_i^j\|_1$, where $\mathbf{x}$ are joint positions computed via forward kinematics on SMPL.
- **GMM NLL**: $\mathcal{L}_{\text{gmm}}$ is a winner-takes-all Gaussian mixture NLL in the 6D rotation space.

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + 0.5\,\mathcal{L}_{\text{vel}} + w_{\text{geo}}\,\mathcal{L}_{\text{geo}} + w_{\text{mpjpe}}\,\mathcal{L}_{\text{mpjpe}} + w_{\text{gmm}}\,\mathcal{L}_{\text{gmm}}, \tag{2}$$

where we ablate the weights $w_\star \in \{0, 0.05, 0.1, 0.2\}$ in our experiments. We found that any weight ≥ 1.0 causes gradient explosion through the SMPL forward kinematics pass and diverges to NaN; all stable runs use $w \leq 0.2$.

3.3 Cross-Attention Pose Encoders (Ablation)

To isolate the contribution of temporal ordering from sequential integration, we test two cross-attention pose encoder variants. In both, the MTR agent feature acts as a single query vector (projected to 256D) and the 11 pose frames act as keys/values (each 144D projected to 256D). The attended output is concatenated with the agent feature.

H5 (no positional encoding). No temporal information is injected before attention. This treats pose frames as a permutation-invariant set and tests whether the raw content of each pose frame—absent any ordering—is sufficient to condition trajectory prediction.

H5b (sinusoidal positional encoding). Standard sinusoidal positional encoding (Vaswani et al., 2017) is added to the projected pose keys/values before attention, injecting the frame index $t \in \{0, \ldots, 10\}$ as fixed (non-learned) sinusoidal features. This adds temporal ordering at zero extra parameter cost and tests whether ordering alone is sufficient, without the causal accumulation structure of a GRU.

4 Experiments

4.1 Dataset and Setup

Dataset. We use a pedestrian-only subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with SMPL body poses from AMASS (Mahmood et al., 2019). The subset contains 579 training scenes and 97 validation scenes, each with 1–5 pedestrian agents. Ground-truth future trajectories come from Waymo tracking data (91 timesteps at 10Hz; 8.1s total). Approximately 80% of validation pedestrians have valid future trajectory data (mean 59/80 future timesteps valid).

Evaluation. We report **minADE** (minimum Average Displacement Error over 6 modes, in meters) and **minFDE** (minimum Final Displacement Error). Lower is better. The evaluation covers the full 8.1s future horizon.

Training. All models train for 30 epochs with batch size 10 on an RTX 4090. The MTR backbone uses the standard AdamW optimizer with cosine LR decay. Training takes approximately 80 seconds per epoch. We report the best validation minADE across all checkpoints.

Limitations of scale. Our 579-scene subset is a small fraction of the full Waymo training set ($\approx$486k scenarios). Consequently, our absolute minADE ($\approx$0.62) is much higher than SOTA on the full dataset ($\approx$0.30). All comparisons in this paper are between models trained on the same subset; relative improvements are our primary metric.

4.2 Main Result: Pose Conditioning vs. Baseline

Table 1 compares the trajectory-only MTR baseline (H2, no pose conditioning) against MTR+Pose with geodesic-only supervision (H3 geo_only), our best configuration.

Table 1: Main result: trajectory-only baseline vs. pose-conditioned MTR+POSE. All models trained 30 epochs on the same 579-scene subset. Δ is relative change vs. baseline.

Model	Pose Enc.	Loss	minADE ↓	minFDE ↓
MTR (no pose)	—	cls, reg, vel	0.6745	1.5080
MTR+POSE (geo only)	GRU	+ geo (w=0.1)	**0.6231**	**1.3982**
Δ (pose vs. no-pose)			−7.6%	−7.3%

Table 2: Ablation of pose supervision signal. All runs: GRU encoder, $w = 0.1$ for active losses, 30 epochs, same data.

Pose Loss	minADE ↓	Δ vs. baseline	Δ vs. geo_only
None (no pose)	0.6745	—	–
MPJPE only	0.6576	−2.5%	+5.5%
GMM NLL only	0.6467	−4.1%	+3.8%
Full (MPJPE+Geo+GMM+Cls)	0.6532	−3.2%	+4.8%
Geodesic only	**0.6231**	−7.6%	—

Pose conditioning with geodesic supervision reduces minADE from 0.6745 to 0.6231, a **7.6% improvement**. This improvement is consistent at minFDE (7.3%).

4.3 ABLATION: POSE SUPERVISION SIGNAL

Table 2 ablates the contribution of each pose loss component. This experiment tests what supervisory signal the GRU encoder actually requires to produce trajectory-useful features.

Three key observations: (1) All pose losses improve over no-pose. (2) Geodesic-only *outperforms the full model*: combining MPJPE with geodesic hurts (0.6532 vs. 0.6231). (3) GMM NLL alone (0.6467) outperforms MPJPE alone (0.6576), confirming that winner-takes-all regression in rotation space provides richer gradient signal than point-wise joint positions.

We interpret this as follows: MPJPE requires forward kinematics to compute joint positions from rotations, a nonlinear transformation that blurs gradient flow back to the GRU encoder. Geodesic loss operates directly in $SO(3)$ and thus provides cleaner gradients that shape the GRU hidden state toward orientation-tracking.

4.4 ABLATION: GEODESIC LOSS WEIGHT

Figure 2 shows the sensitivity of the geodesic-only configuration to the loss weight w_{geo}.

The optimal weight is **sharply** at $w = 0.1$: both $w = 0.05$ (0.6786) and $w = 0.2$ (0.6660) are significantly worse, nearly matching the no-pose baseline. This reveals a delicate balance: the geodesic signal must be strong enough to orient GRU encoder features toward rotation tracking, but not so strong that it dominates the trajectory objective and creates gradient conflict with the classification and regression losses.

4.5 ABLATION: TEMPORAL ENCODING — GRU VS. CROSS-ATTENTION

Table 3 compares three temporal encoding strategies, all with geodesic-only supervision at $w_{\mathrm{geo}} = 0.1$.

Cross-attention without positional encoding gives 0.6765—no improvement over baseline—because it treats the 11 pose frames as a permutation-invariant set. Adding sinusoidal positional encoding (cross-attention + PE) recovers **83%** of the GRU's advantage, reaching minADE = 0.6337 (-6.1%). The GRU achieves 0.6231 (-7.6%), a further 1.7% improvement over cross-attention with PE.

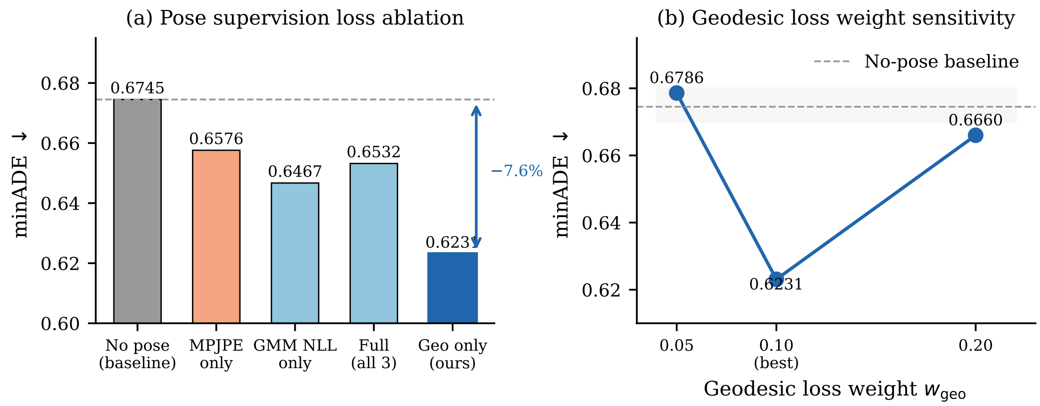

Figure 2: **Loss ablation and geodesic weight sensitivity.** *Left*: minADE for five pose supervision configurations. Geodesic-only supervision achieves the best result (0.6231); combining MPJPE with geodesic hurts relative to geodesic alone. *Right*: Sensitivity of the geodesic-only model to the loss weight w_{geo}. The optimum is sharp at $w = 0.1$; both $w = 0.05$ and $w = 0.2$ nearly match the no-pose baseline. Dashed line marks the no-pose baseline (0.6745).

Table 3: Temporal encoding ablation. All pose-conditioned variants use geodesic-only supervision ($w = 0.1$).

Pose Encoder	Temporal info	minADE ↓	Δ vs. baseline
None (no pose)	—	0.6745	—
Cross-attention, no PE	none (bag)	0.6765	+0.3%
Cross-attention + sinus. PE	ordering only	0.6337	−6.1%
GRU (sequential)	ordering + causal	**0.6231**	−7.6%

The progression reveals a clean decomposition: (1) temporal *ordering*—knowing *when* each frame occurred—is the dominant requirement, contributing 83% of the total gain. Sinusoidal PE, which injects position indices without learned weights, is sufficient to capture this. (2) The GRU's causal *sequential integration*—where each hidden state is a running summary of all preceding frames—provides an incremental further gain. Cross-attention with PE attends to all frames simultaneously; the GRU accumulates a compressed history that may be more compatible with the trajectory decoder's expectation of a single context vector. Geodesic supervision requires temporal context to shape encoder features—without ordering (bag-of-items), the geodesic gradient cannot encode gait dynamics.

4.6 Ablation: Future Pose Supervision Quality

To localize the source of improvement, we compare two data conditions: (a) *no future poses*: future pose GT is zero, only past poses are observed (our main experimental condition); and (b) *30fps AMASS future poses*: AMASS motion sequences provide 99% coverage of future timesteps as pseudo-GT.

The result: the 30fps model achieves minADE $= 0.6567$, and the zero-future model achieves 0.6532—a difference of 0.0035 (0.5%), within noise. **Better future pose supervision does not improve trajectory prediction**. This localizes the entire 7.6% gain to the past GRU encoder: encoding how the body has moved up to the present provides useful trajectory context; predicting where the body will go provides no additional benefit.

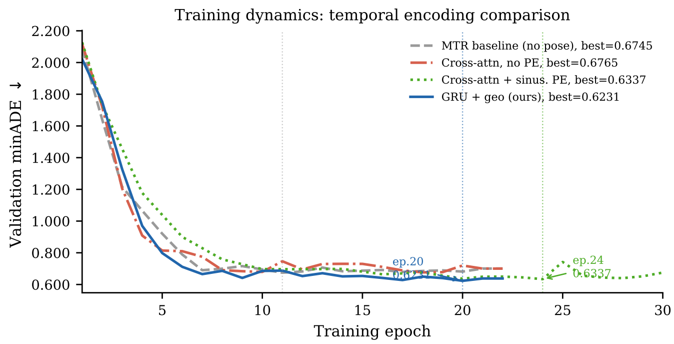

Figure 3: **Training dynamics.** Validation minADE for all four temporal encoding variants. Cross-attention without PE (orange, dash-dot) tracks the baseline throughout, confirming that permutation-invariant aggregation extracts no temporal information. Cross-attention with sinusoidal PE (green, dotted) drops to 0.6337 at epoch 24, recovering 83% of the GRU's advantage. GRU+geo (blue, solid) reaches 0.6231 at epoch 20 with higher epoch-to-epoch variance, consistent with the additional SMPL pose matching signal.

4.7 Training Dynamics

Figure 3 shows validation minADE over 30 training epochs for all four temporal encoding variants. Cross-attention without PE (orange) closely tracks the trajectory-only baseline, visually confirming the null result. Adding sinusoidal PE (green) produces a qualitatively different trajectory, eventually reaching 0.6337 at epoch 24. MTR+Pose with GRU+geo (blue) reaches its best of 0.6231 at epoch 20 with higher epoch-to-epoch variance, consistent with the additional noise from the SMPL pose supervision signal.

5 Analysis and Insights

Why does geodesic loss outperform MPJPE? SMPL joint positions are computed via a nonlinear kinematic chain: $\mathbf{x}_j = \mathrm{FK}(\theta_1, \ldots, \theta_j)$, where $\theta_k \in SO(3)$. Gradient of $\mathcal{L}_{\text{mpjpe}}$ w.r.t. the GRU hidden state traverses this kinematic chain, introducing Jacobians that can be small or misaligned with the gait-relevant rotation axes. Geodesic loss operates directly on $\mathrm{tr}(\mathbf{R}^\top \hat{\mathbf{R}})$, a smooth function of rotation matrices, providing gradients that are proportional to the angular error without kinematic compounding.

What makes the GRU encoder work? The temporal ablation decomposes the GRU's advantage into two separable components. *First*, temporal ordering—knowing *when* each frame occurred—is the dominant requirement. Sinusoidal positional encoding, a fixed injection of position indices requiring no learned parameters, is sufficient to recover 83% of the GRU's gain (-6.1% vs. -7.6%). This is consistent with gait being primarily a phase-ordered phenomenon: knowing that a left-foot strike at $t = 3$ follows right-foot strike at $t = 1$ captures most of the useful temporal structure. *Second*, the GRU's causal *sequential integration* adds an incremental 1.7%. The GRU hidden state $\mathbf{h}_T$ is a compressed running summary of all preceding frames; cross-attention with PE attends to all 11 frames simultaneously and may produce a representation less compatible with the trajectory decoder's expectation of a single compact context vector. With geodesic supervision shaping $\mathbf{h}_T$ toward rotation tracking, the sequential accumulation captures gait phase transitions more efficiently than global self-attention.

The mechanism: past encoder, not future prediction. The null result of our future-pose ablation (Section 4.6) is strong evidence that the mechanism is *pose encoding*, not *pose prediction*. The GRU acts as a feature extractor for the MTR decoder, providing an orientation-aware agent representation that improves trajectory intention localization. The pose decoder (future prediction head) is an auxiliary task that supervises the encoder but does not itself contribute to trajectory accuracy.

6 LIMITATIONS

Dataset scale. Our experiments use 579 training scenes from Waymo, a small fraction of the full dataset. While relative comparisons are internally valid, we cannot claim that the 7.6% improvement transfers directly to a full-scale training regime.

SMPL pose availability. We use AMASS-derived pose sequences matched to Waymo pedestrian tracks, which requires a separate pose estimation step at test time. In deployment, online body pose estimation (e.g., from video) introduces estimation noise not present in our evaluation. The sensitivity of our method to noisy pose inputs is not evaluated here.

No HD map features. The SMPL-paired subset lacks HD map polylines. We use zero-padded map features, which means our model does not benefit from map-aware pedestrian routing. Results on map-equipped data may differ.

Pose decoder quality. We do not evaluate MPJPE/geodesic error of the predicted future poses independently; our metric is trajectory prediction only. A joint evaluation of pose and trajectory quality would complete the picture.

Single-seed results. All 14 runs use a single random seed. Epoch-to-epoch variance in our training curves (particularly the LR-decay spike at epoch 25 in the H5b run, where minADE jumps to 0.7420 before recovering) suggests non-trivial training noise. We report best-checkpoint minADE, which may slightly overfit to the noise of a single run. Repeating key runs with multiple seeds would tighten the confidence interval on the reported improvements.

7 CONCLUSION

We have presented MTR+POSE, a pose-conditioned extension of the Motion Transformer that incorporates past SMPL body poses via a GRU encoder with 6D rotation representations. Through a systematic ablation of 14 runs on the Waymo Open Motion Dataset, we establish that: (1) geodesic supervision in $SO(3)$ is the critical ingredient, reducing minADE by 7.6% over the trajectory-only baseline, while joint-space MPJPE supervision yields only 2.5% improvement and degrades performance when combined with geodesic loss; (2) temporal ordering of pose frames is the dominant requirement—cross-attention *without* positional encoding provides no improvement (0.6765), while adding sinusoidal PE recovers 83% of the GRU's advantage (0.6337, -6.1%); the GRU's causal sequential integration provides a further incremental gain to -7.6%; (3) the source of improvement is the past pose encoder, not future pose prediction quality.

These findings suggest that rotation-space supervision and temporal ordering are underappreciated design choices in pose-conditioned forecasting. We hope the systematic ablation methodology contributes to more principled design of multi-modal pedestrian prediction systems.

ACKNOWLEDGMENTS

The authors gratefully acknowledge access to GPU compute (RTX 4090) and the publicly available Waymo Open Motion Dataset.

REFERENCES

Ling-Hao Chen, Jiawei Zhang, Yewen Li, Yiren Pang, Xiaobo Xia, and Tongliang Liu. HumanMAC: Masked motion completion for human motion prediction. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9510–9521. IEEE, October 2023. doi: 10.1109/iccv51070.2023.00875. URL `http://dx.doi.org/10.1109/iccv51070.2023.00875`.

Scott Ettinger, Shuyang Cheng, Benjamin Caine, Chenxi Liu, Hang Zhao, Sabeek Pradhan, Yuning Chai, Ben Sapp, Charles Qi, Yin Zhou, Zoey Yang, Aurelien Chouard, Pei Sun, Jiquan Ngiam, Vijay Vasudevan, Alexander McCauley, Jonathon Shlens, and Dragomir Anguelov. Large scale interactive motion forecasting for autonomous driving : The waymo open motion dataset. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9690–9699. IEEE, October 2021. doi: 10.1109/iccv48922.2021.00957. URL `http://dx.doi.org/10.1109/iccv48922.2021.00957`.

Agrim Gupta, Justin Johnson, Li Fei-Fei, Silvio Savarese, and Alexandre Alahi. Social GAN: Socially acceptable trajectories with generative adversarial networks. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2255–2264. IEEE, June 2018. doi: 10.1109/cvpr.2018.00240. URL http://dx.doi.org/10.1109/CVPR.2018.00240.

Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: a skinned multi-person linear model. *ACM Transactions on Graphics*, 34(6):1–16, October 2015. ISSN 1557-7368. doi: 10.1145/2816795.2818013. URL http://dx.doi.org/10.1145/2816795.2818013.

Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael Black. AMASS: Archive of motion capture as surface shapes. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 5441–5450. IEEE, October 2019. doi: 10.1109/iccv.2019.00554. URL http://dx.doi.org/10.1109/iccv.2019.00554.

Tim Salzmann, Boris Ivanovic, Punarjay Chakravarty, and Marco Pavone. *Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data*, pp. 683–700. Lecture Notes in Computer Science. Springer International Publishing, 2020. ISBN 9783030585235. doi: 10.1007/978-3-030-58523-5_40. URL http://dx.doi.org/10.1007/978-3-030-58523-5_40.

Shaoshuai Shi, Li Jiang, Dengxin Dai, and Bernt Schiele. Motion transformer with global intention localization and local movement refinement. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 6531–6543. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/2ab47c960bfee4f86dfc362f26ad066a-Paper-Conference.pdf.

Shaoshuai Shi, Li Jiang, Dengxin Dai, and Bernt Schiele. MTR++: Multi-agent motion prediction with symmetric scene modeling and guided intention querying. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3955–3971, May 2024. ISSN 1939-3539. doi: 10.1109/tpami.2024.3352811. URL http://dx.doi.org/10.1109/tpami.2024.3352811.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Ye Yuan, Xinshuo Weng, Yanglan Ou, and Kris Kitani. Agentformer: Agent-aware transformers for socio-temporal multi-agent forecasting. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9793–9803. IEEE, October 2021. doi: 10.1109/iccv48922.2021.00967. URL http://dx.doi.org/10.1109/iccv48922.2021.00967.

Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5738–5746. IEEE, June 2019. doi: 10.1109/cvpr.2019.00589. URL http://dx.doi.org/10.1109/CVPR.2019.00589.

Wentao Zhu, Xiaoxuan Ma, Zhaoyang Liu, Libin Liu, Wayne Wu, and Yizhou Wang. MotionBERT: A unified perspective on learning human motion representations. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 15039–15053. IEEE, October 2023. doi: 10.1109/iccv51070.2023.01385. URL http://dx.doi.org/10.1109/ICCV51070.2023.01385.

A Additional Experimental Details

A.1 Architecture Hyperparameters

- GRU hidden size: 256

- Pose input dimension: 144 (24 joints × 6D rotation)
- Past pose window: 11 timesteps (1.0 second at 10 Hz)
- Number of predicted trajectory modes: $K = 6$
- Number of future steps: 80 (8.0 seconds)
- MTR encoder: 4-layer transformer, 256 hidden dim, 8 heads
- Batch size: 10
- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$
- LR schedule: Cosine decay, initial LR 1×10^{-3}
- Epochs: 30
- Gradient clipping: norm ≤ 1.0
- Dropout: none (small dataset)

A.2 Dataset Construction

Waymo agent future trajectories are stored in `processed_scenarios/` PKL files with shape `(num_agents, 91, 10)`. SMPL pedestrian files are matched to Waymo object IDs via integer stem matching. Approximately 80% of validation pedestrians have a valid trajectory match; unmatched pedestrians are excluded from evaluation. The 91-step grid spans $[0, 9.0]$ seconds; SMPL timestamps (originally in microseconds) are divided by 10^6 before alignment.

A.3 Loss Weight Stability

At $w = 1.0$, gradient explosion through the SMPL forward kinematics causes NaN divergence at epoch 3. The geodesic loss gradient norm was also found to diverge at the ±1 boundary of the $\arccos$ argument; we clamp to $[-1+\epsilon, 1-\epsilon]$ with $\epsilon = 10^{-7}$. At $w = 0.1$, training is stable for all 30 epochs across all ablation variants.

A.4 Full Results Table

Table 4 reports all 14 experimental runs including invalid (zero future GT) baselines for completeness.

Table 4: Complete experimental results. Runs 001–002 are invalid (zero future GT) and excluded from analysis. Baseline for Δ is run 004.

Run	Tag	Encoder	Losses (w)	minADE	Δ
001	H1 full pose	GRU	all (1.0)	0.6114	*invalid*
002	H2 no pose	—	traj only	0.7724	*invalid*
003	H1_v2 full	GRU	all (0.1)	0.6532	−3.2%
004	H2_v2 baseline	—	traj only	0.6745	—
005	H1 w=0.05	GRU	all (0.05)	0.6557	−2.8%
006	H1 w=0.2	GRU	all (0.2)	0.6542	−3.0%
007	H1_v3 30fps	GRU	all (0.1), 30fps fut	0.6567	−2.6%
008	H3 mpjpe only	GRU	mpjpe+gmm (0.1)	0.6576	−2.5%
009	H3 gmm only	GRU	gmm (0.1)	0.6467	−4.1%
010	H3 geo only	GRU	geo+gmm (0.1)	**0.6231**	−7.6%
011	H3 geo w=0.05	GRU	geo+gmm (0.05)	0.6786	−1.5%
012	H3 geo w=0.2	GRU	geo+gmm (0.2)	0.6660	−1.3%
013	H5 cross-attn	XAttn	geo+gmm (0.1)	0.6765	+0.3%
014	H5b xattn+PE	XAttn+PE	geo+gmm (0.1)	0.6337	−6.1%